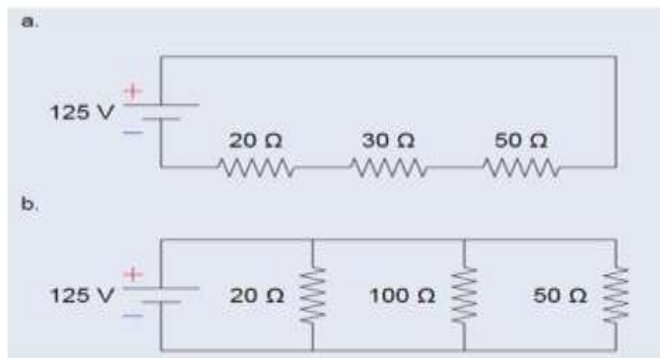


Embedded Systems & IoT
Lab Exercise: Schematics with ESP32
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Part A



Circuit 1 – Circuit is in series

a. Total Resistance

$$\begin{aligned}\text{Total Resistance} &= \text{Sum of Individual Resistance} \\ &= 20\Omega + 30\Omega + 50\Omega \\ &= \mathbf{100\Omega}\end{aligned}$$

b. Total Current from the power supply

Using Ohm's Law:

$$I = V/R \quad \text{where } V = \text{Voltage, } I \text{ is current and } R = \text{Resistance}$$

$$I = 125/100$$

$$I = \mathbf{1.25\Omega}$$

c. Current through each resistor

Since this circuit is in series, current through each resistor is the same, therefore it is **1.25A**

d. Voltage Drop Across each resistor

Reworking Ohm's Law;

$$V = IR$$

Therefore, for 20 Ω:

$$V = 1.25 * 20 = 25V$$

For 30Ω:

$$V = 1.25 * 30 = 37.5V$$

For 50Ω:

$$V = 1.25 * 50 = 62.5V$$

e. Power Dissipated through each resistor

Using:

$$P = VI$$

For 20 Ω:

$$P = 25 * 1.25 = 31.25W$$

For 30 Ω:

$$P = 37.5 * 1.25 = 46.875W$$

For 50 Ω:

$$P = 62.5 * 1.25 = 78.125W$$

Circuit 2 – Parallel Circuit

1. Total Resistance

Since this is a parallel circuit:

$$\begin{aligned} 1/R(T) &= 1/20 + 1/100 + 1/50 \text{ where } R(T) \text{ is the total resistance} \\ &= 0.05 + 0.01 + 0.02 \\ &= 0.08 \end{aligned}$$

Therefore, Total Resistance = $1/0.08 = \mathbf{12.5 \Omega}$

2. Total Current from the Supply

$I(T) = V/R(T)$ where I is the total current, V is the voltage and R is the total resistance

$$\begin{aligned} I(T) &= 125/12.5 \\ I(T) &= \mathbf{10A} \end{aligned}$$

3. Current through each resistor

Since this is a parallel circuit, voltage across each resistor is the same:

$$V = 125V$$

Using:

$$I = V/R$$

20 Ω :

$$I = 125/20$$

$$\mathbf{I = 6.25A}$$

100 Ω

$$I = 125/100$$

$$\mathbf{I = 1.25A}$$

50 Ω

$$I = 125/50$$

$$\mathbf{I = 2.5A}$$

4. Voltage drop across each resistor

Since the circuit is in parallel, voltage across each resistor is the same which will thus be **125V**

5. Power dissipated through each resistor

$P=VI$ where P =Power, V = Voltage and I = Current

20 Ω

$$P=125*6.25= 781.25W$$

100 Ω

$$P=125*1.25= 156.25W$$

50 Ω

$$P=125*2.5=312.5W$$